

Himanshu Singhal

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EDUCATION

Indian Institute of Technology Guwahati, India

Bachelors of Technology in Data Science and Artificial Intelligence

Oct 2022 - Present

GPA: 8.10/10.0

EXPERIENCE

Adobe Research

Summer Intern, Internship

May 2025 - Aug 2025

Bangalore, India

- Designed and trained **diffusion-based generative models** (Diffusion Transformers) for controllable visual asset generation under strict conditioning constraints
- Built **multimodal conditioning pipelines** mapping structured inputs (hex color codes) to embedding space for precise controllability in generation
- Developed **data curation and evaluation pipelines**, including a **1200-prompt benchmark suite** for systematic regression testing
- Improved generation fidelity by **64%** through architectural and conditioning improvements
- Filed and secured internal approval for a **patent** covering the end-to-end controllable generation pipeline

Online Signature Verification using Vision Transformers

BTech Project, Prof. Suresh Sundaram

Jul 2025 - Present

Guwahati, India

- Investigating **online signature verification** using **Vision Transformers** to better model temporal dependencies and represent fine-grained signature dynamics, leading to improved authentication accuracy.
- Reformulated signature verification as a **one-class anomaly detection** problem, allowing the model to be trained solely on genuine samples without requiring forged data for supervision.
- Modeled temporal signature signals as **image-based representations** using **Gramian Angular Summation Field (GASF)** plots to capture fine-grained dynamics while mitigating temporal distortions and local misalignments.
- Evaluated the proposed framework on four public benchmarks—**BioSecureID**, **SVC2021**, **DeepSignDB**, and **MCYT-100**—achieving performance competitive with state-of-the-art methods.

Arizona State University - CORAL Lab

Research Intern, Prof. Vivek Gupta

Feb 2025 - Present

Remote

- Built an **end-to-end ML pipeline** for multi-table query generation, including data preprocessing, retrieval, and structured output generation
- Designed efficient retrieval systems using **inter-table join graphs** for large-scale structured datasets
- Reduced token usage by **92.24%** and improved inference latency by **4.6×** through system-level optimizations
- Developed **evaluation pipelines** for benchmarking against LLM-based baselines
- Paper **accepted to ACL 2026 (Main Conference)**, placing in the **top 5% of submissions globally**

SustainAI Lab - IITG

Research Intern, Prof. Ayon Borthakur

Jan 2025 - May 2025

Guwahati, India

- Implemented and trained **memory-augmented architectures** (Titans) under constrained compute settings
- Built **evaluation pipelines** using LM-Eval for systematic benchmarking
- Explored **hybrid memory-attention architectures** combining persistent memory with attention mechanisms

Devrev Cloud India Private Limited

Software Intern, Internship

Feb 2024 - Apr 2024

Remote

- Developed a **customizable NPS survey snap-in** for developer organizations, enabling them to dispatch surveys at selected intervals, streamlining feedback collection, and boosting user engagement.
- Utilized **hooks** and **cron jobs** to facilitate sending emails at specified intervals, and employed frameworks such as **React-Email** to guarantee compatibility and a consistent user experience across different email clients.
- Built **real-time** survey update mechanisms and **interval-based** email dispatch system ensuring consistency.
- Deployed an automated survey distribution system serving **1,000+ users**, improving feedback collection efficiency.

PUBLICATIONS

REaR: Retrieve, Expand and Refine for Effective Multitable Retrieval.

R. Agarwal*, **H. Singhal*** P.B. Chen, M.R. Choudhury, D. Roth, V. Gupta

ACL 2026 (Main)

- Association for Computational Linguistics (ACL) 2026.
- arXiv: 2511.00805**

PROJECTS

Agentic RAG Chatbot for LegalQA Inter IIT Tech Meet 13.0	Oct 2024 - Dec 2024 Github
- Built a production-grade RAG pipeline with real-time document ingestion and retrieval using Pathway - Designed retrieval pipelines with ranking and reranking for improved context selection - Implemented DAG-based execution pipelines for parallel task orchestration - Achieved 86% accuracy on CUAD and 84% on AILA with optimized latency and cost	
Adobe Behaviour Simulation Challenge Inter IIT Tech Meet 12.0	Oct 2023 - Dec 2023 Github
- Leveraged classic ML Models, Transformers, and LLMs for behaviour and content simulation on tweets. - Trained a DistilBERT classifier to bucket tweets, followed by separate bucket regressors for likes prediction. - Generated descriptions of media using BLIP-2 and combined with metadata to finetune Llama-2 for captioning. - Achieved a bucket classification accuracy of 74.3% and BLEU-1 score of 0.13 with inference times 0.9s and 5.4s.	
Devrev's AI Agent 007 Inter IIT Tech Meet 12.0	Oct 2023 - Dec 2023 Github
- Engineered a conversational LLM chatbot with dynamic tool selection capabilities for intelligent query resolution. - The chatbot selects and sequences appropriate tools with required arguments to accurately answer user queries. - Implemented a pipeline integrating Retrieval-ICL for few-shot retrieval, Chain-of-Thought prompting, corrective reprompting, and RAG with Dynamic Retrieval to bolster the model's contextual understanding. - Achieved high-performance pipeline with sub-10-second query latency and cost efficiency under \$0.02 per query.	
Other Projects Telematics Data Analysis : Data pipeline with real-time Kafka streaming, risk prediction, and visualisation dashboard Interview Chatbot : Chatbot with Google Vertex AI for stimulated interview and feedback with Codeforces API	

ACHIEVEMENTS

Adobe Research	Patent application authored from internship research and approved through internal review	2025
Inter IIT Tech Meet 13.0	Secured Rank 4 among all 23 IITs in ML/AI Problem Statement by Pathway	2024
Encode, IITG	Team secured first position out of 2342 participants at ML hackathon across India	2024
Amazon MLSS	Selected to attend ML lectures from more than 20,000 students in India .	2024

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, TypeScript	Systems & Infra: REST APIs, Docker, real-time inference pipelines
ML / DL: PyTorch, Transformers, Diffusion Models, Vision Transformers, Multimodal Models	Model Optimization: Latency optimization, batching, evaluation pipelines
Data & Pipelines: PySpark, Pandas, NumPy, Kafka, large-scale data processing, data curation	Tools: Git, Shell, LangChain, LangGraph, LlamaIndex

RELEVANT COURSEWORK

Computer Science: Data Structures and Algorithms (Theory & Lab), Database Management Systems (Theory & Lab), Computer Systems, Big Data Analytics, Machine Learning, Deep Learning, Data Mining, Advanced Deep Learning
Mathematics: Linear Algebra, Statistical Foundations of Data Science, Applied Probability & Random Processes

POSITIONS OF RESPONSIBILITY

- Teaching Assistant, DA205**, Responsible for curating weekly exam question banks and supporting live sessions.
- Secretary, IITG.ai**, Responsible for managing events, workshops, competitions, and discussions at IITG.ai club, the AI Community of IIT Guwahati.